ON SERIES OF BESSEL FUNCTIONS

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$$\Sigma V_r = \Sigma A_r \int_z^1 z^{-m+1} \left\{ \int_0^z t^{\frac{1}{2}m} J_m\{k_z \sqrt{t}\}\, dt \right\} dz$$

converges uniformly to the function

$$\psi(z) = \int_z^1 z^{-m-1} \left\{ \int_0^z t^{\frac{1}{2}m} f(\sqrt{t})\, dt \right\} dz,$$

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1. The theory of these series as hitherto exposed may be regarded as involving serious difficulties of analysis. Till quite recently Dini's work constituted the last word on the subject. Previously a long series of mathematicians had occupied themselves with the question, including names like Fourier, Hankel, and Harnack. Yet Dini seems to have been

the first to obtain reliable results. The conditions due to him are somewhat complicated in character, and, in the light of recent advances in Analysis, more especially in the theory of Fourier series, far from final. Kneser has availed himself of the modern subject of Integral Equations to obtain results of an isolated nature for Bessel series $\Sigma A_r J_m(k_r x)$, where

$$A_r = C_r \int_0^1 J_m(k_r x)\, x f(x)\, dx,$$

for the case $m = 0$, and Hobson, basing himself, as he tells us, partly on Kneser's work, has proved that the Bessel series of a function $f(x)$ converges at every point of the half-closed interval $(0 < x < 1)$ at which the function has bounded variation, provided only that $\sqrt{x}\, f(x)$ is summable.

I have already, in a paper dealing with series of Legendre's Polynomials, explained a general method of treating the expansion of $f(x)$ in a series of functions, based on a fundamental theorem which I have recently discovered in the theory of what I have called Restricted Fourier Series. The application to Legendre's polynomials exhibits the method in a peculiarly pure light, as the asymptotic approximation for $P_n(x)$ involves trigonometrical functions of the type employed in harmonic trigonometrical series, viz. $\sin nx$ and $\cos nx$, where n is an integer. Moreover the fact that the functions in terms of which we expand are polynomials enables us to avoid entirely the use of the complex variable.

In order then to exhibit the full power of the method, it has seemed to me desirable to discuss a case where these simplifications do not present themselves. I have chosen the theory of series of Bessel functions $\Sigma A_r J_m(k_r x)$ with this object in view, and the more readily because, as has been just pointed out, so little has hitherto been achieved in the theory of these series.

I have been at pains to explain the processes employed as carefully as possible, so as to render them intelligible to those who are unacquainted with previous work on the subject. There is indeed little connexion between the work here exposed and that of previous authors.

As regards the conclusions arrived at, and indicated, these may, I think, fairly be said to be of a far-reaching nature, and constitute little less than the complete solution of the problem of determining the behaviour of the series under consideration in the matter of convergence, uniform convergence, convergence by the methods of Cesàro, oscillation, and so forth, at any *internal* point of the interval (0, 1). Thus it is shown that, *under the sole condition—necessary for convergence—that the typical term of a Bessel series tends to zero, the series behaves at any such internal*

point precisely like a Fourier series, and we may obtain, without difficulty, circumstances under which it will converge in the Cesàro manner, from a knowledge of the power to which we must raise the inverse of the index expressing position in the series in order that the typical term, multiplied by this power, should tend to zero. On the other hand very general sufficient conditions, which the function $f(x)$ must satisfy in order that the coefficients may be of the required order, can be at once written down.

One of the more interesting features of the discussion is perhaps the introduction of trigonometrical series of the non-harmonic type

$$\Sigma \{a_n \cos(k+n)x + b_n \sin(k+n)x\},$$

which are shown to have properties analogous to those possessed by ordinary trigonometrical series, equally whether k be rational or irrational.

Again, it is of importance to remark that the scope of the investigation does not only include series of the type hitherto studied (*Bessel series*) in which the coefficients are expressible by means of integrals involving the function $f(x)$. Our method gives results of scarcely less importance when the coefficients are not of this form (*series of Bessel functions*), in accordance with the analogy suggested by the step taken in passing from ordinary Fourier series to Restricted Fourier Series. In particular we are able to deal with the derived series of Bessel series, although such derived series in our case necessarily cease to have the same form as the series from which they are derived. We are thus able to complete in a notable point the usual investigations in the applications. In fact, *it has been usual to assume tacitly that, if the individual term satisfies a certain differential equation, the function represented by the sum of an infinite series of such terms will also satisfy it. It is here shewn that such a conclusion is legitimate, if the function $f(x)$ have, in the interval in which such an equation is to be satisfied, a differential coefficient of suitable order, such that the Fourier series of a function equal to this differential coefficient in this interval, and having any convenient values elsewhere, converges in the interval in question. This is, for instance, the case if the differential coefficient has bounded variation.*

It* will be seen that our method, based as it is on the asymptotic trigonometric approximation [(viii) of § 4], which only holds in an interval excluding the origin, gives us no information as to what happens at $z = 0$. When $m > 0$, the series obviously converges there to zero, but

* Added January 29th, 1918.

it remains an open question as to the conditions under which the convergence is bounded in the neighbourhood, or, in the case when the function approaches zero as limit there, uniform. On the other hand we are able, by considerations analogous to those employed below, to deal with the end-point $z = 1$, obtaining a result exactly corresponding to that for an internal point of the interval (0, 1). The exposition of these latter considerations involves, however, a more detailed treatment of the theory of non-harmonic trigonometrical series than has been found possible to include in the present communication.* The question, moreover, of convergence of a series, at the extremity of an interval in which it defines an associated function, requires and deserves a fuller and separate discussion, and I propose shortly to make a communication to the Society on these and kindred matters.

On the other hand, in view of the fact that I have not found it convenient to dispense entirely with the use of the complex variable, I have thought it well to shew in an Appendix what additional considerations, drawn from the theory of the complex variable alone, and avoiding, in particular, the use of the theory of Integral Equations, suffice for the proof of the known theorem that a Bessel series converges at $z = 1$, when its associated function $f(z)$ has bounded variation there and $\sqrt{z}\, f(z)$ is summable.

It should finally be remarked that the results of the paper hold as they stand, not only for all values of $m \geqslant 0$, but also for $m \geqslant -\frac{1}{2}$, and even for all values of m greater than -1, provided slight modifications be made, involving a change in the condition of summability in the neighbourhood of the origin of the function $f(z)$ associated with the series.

2. The series which we are going to investigate are *series of Bessel functions*

$$\Sigma A_r J_m(k_r z), \tag{1}$$

where m is a fixed positive quantity (or zero), and the numbers $k_1, k_2, \ldots, k_r, \ldots$ form a monotone increasing succession tending to infinity, and are deter-

* From the form of the first approximation given in (25a) of § 25, these conditions of convergence can be deduced. We have, in fact, a series of cosines of $(k + r)\, z$, where $\pi z = \pi - y$, so that there is symmetry about the origin of y. It then follows from a general theorem, given by me in a recent communication to the Royal Society, that this non-harmonic trigonometrical series behaves in respect of convergence and uniform convergence at $y = 0$, exactly like the expansion in such a series of a function which is summable in the interval $(0, 2\pi)$, and, at and in the neighbourhood of the origin $y = 0$, is equal to $f(1 - y/\pi)$. By the theory of such expansions we then perceive the truth of the statement made in the text.

mined as the positive roots of the transcendental equation

$$kJ'_m(k)+HJ_m(k)=0, \tag{2}$$

involving a parameter H at our disposal.* We may accordingly write

$$k_r\pi=\tfrac{1}{2}m+\tfrac{1}{4}+s+r+r^{-1}a(r), \tag{3}$$

where s is a certain fixed integer, and $a(r)$ is a bounded function of r of the form

$$a(r)=C+r^{-1}a_1(r),$$

$a_1(r)$ being another bounded function of r.

3. The series (1) is said to be *a Bessel series*, if the coefficients A_r have the form

$$A_r=C_r\int_0^1 J_m(k_rt)\,tf(t)\,dt=\tfrac{1}{2}C_r\int_0^1 f\sqrt{t}\,J_m(k_r\sqrt{t})\,dt, \tag{4}$$

where C_r is given by the equations

$$\tfrac{1}{2}C_r=1\Big/\int_0^1\{J_m(k_r\sqrt{t})\}^2\,dt=1\Big/\left\{\left(1-\frac{m^2}{k_r^2}\right)J_m^2(k_r)+J_m'^2(k_r)\right\}$$

$$=k_r^2/\{H^2+k_r^2-m^2\}\,J_m^2(k_r)=O(k_r). \tag{5}$$

4. I recall the definition of Bessel's function $J_m(x)$,†

$$J_m(x)=\frac{x^m}{2^m\,\Gamma(m+1)}\left\{1-\frac{x^2}{2(2m+2)}+\frac{x^4}{2.4(2m+2)(2m+4)}+\dots\right\},$$

this being valid for all values of x other than real and negative values. The following known properties of $J_m(x)$ are required in the sequel:—

(i) $J_m(x)$ is a bounded function of x in the infinite interval $0\leqslant x$; indeed $\sqrt{x}\,J_m(x)$ is a bounded function of x for $0\leqslant x$;‡ $x^{-m}J_m(x)$ is evidently bounded in any finite interval $(0, B)$;

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* E. W. Hobson, "On the Representation of a Function by a Series of Bessel's Functions", 1908, *Proc. London Math. Soc.*, Ser. 2, Vol. 7, pp. 359–388. In the formula on p. 366,

$$\lambda_r^{\frac{1}{2}}=k_r.$$

† I. Todhunter, *An Elementary Treatise on Laplace's Functions, Lamé's Functions, and Bessel's Functions* (1875), Chapters XXX *et seq.*

‡ Hobson, *loc. cit.*, p. 366.

(ii) the differential equation satisfied by $J_m(x)$ is

$$x\frac{d}{dx}[xJ'_m(x)]-(m^2-x^2)J_m(x)\equiv J''_m+x^{-1}J'_m+(1-m^2x^{-2})J_m=0;$$

(iii) $$mJ_m(x)=\tfrac{1}{2}x\{J_{m-1}(x)+J_{m+1}(x)\};$$

(iv) $$J'_m(x)=\tfrac{1}{2}\{J_{m-1}(x)-J_{m+1}(x)\};$$

(v) $$\int_0^1 J_m(k_r\sqrt{t})\,J_m(k_n\sqrt{t})\,dt=0 \quad (n\neq r)$$

[the value of this integral when $n=r$ is $2/C_r$, as in (5) above];

(vi) $$\int_0^x t^{m+1}J_m(t)\,dt=x^{m+1}J_{m+1}(x);$$

(vii) $$\int_0^x t^{-\frac{1}{2}(m+1)}J_{m+1}(\sqrt{t})\,dt=2\left\{\frac{1}{2^m m}-x^{-\frac{1}{2}m}J_m(\sqrt{x})\right\};$$

(viii) $$\sqrt{(\tfrac{1}{2}\pi x)}\,J_m(x)$$
$$=\cos\psi+\frac{m^2-\frac{1}{4}}{2x}\sin\psi-\frac{(m^2-\frac{1}{4})(m^2-\frac{9}{4})}{2!\,(2x)^2}\left[\cos\psi+\theta\frac{m^2-\frac{16}{4}}{3(2x)}\right],*$$

where $|\theta|\leqslant 1$, and $\psi=-x+\frac{1}{4}(2m+1)\pi$.

From this *asymptotic expression* for $J_m(x)$, we at once get the following, using (iv), and remarking that the change of m into $m-1$ subtracts $\frac{1}{2}\pi$ from ψ, and that into $m+1$ adds $\frac{1}{2}\pi$:—

(ix) $$\frac{d}{dx}\{\sqrt{(\tfrac{1}{2}\pi x)}\,J_m(x)\}$$
$$=\sqrt{(\tfrac{1}{2}\pi x)}\{J'_m(x)+J_m(x)/2x\}$$
$$=\sin\psi-\frac{n^2-\frac{1}{4}}{2x}\cos\psi-\frac{(n^2-\frac{1}{4})(n^2+\frac{7}{4})}{2!\,(2x)^2}\sin\psi+O(x^{-3}).$$

5. In the first instance we are especially interested in those series for which the typical term tends to zero, *i.e.*

$$A_r J_m(k_r z)\to 0 \quad (r\to\infty). \tag{6}$$

This condition is evidently *necessary*, if the series (1) is to converge, and it will be seen that its assumption renders the series as manageable as a

* Lipschitz, *Journal für Math.*, Vol. 56, 1859, pp. 189–196: reproduced in A. Gray and G. B. Mathews, *A Treatise on Bessel Functions*, 1895, p. 70.

Fourier series. It is not satisfied of itself when the series is a Bessel series. In fact the order of magnitude of the typical term varies with the properties of the associated function $f(x)$.

It is immediately evident that no more is required, in order that the usual expression for the coefficients A_r, viz. (4), should have a meaning, than that $x^{m+1}f(x)$ should be absolutely integrable (summable). On the other hand, it is easily seen that the absolute integrability of $\sqrt{x}f(x)$ is sufficient to ensure the convergence of the typical term to zero.

More generally we have the following theorem:—

THEOREM.—*If* $x^{\frac{1}{2}+s}f(x)$ *is absolutely integrable, where* $0 \leqslant s \leqslant m+\frac{1}{2}$, *then*

$$r^{-s}A_r J_m(k_r z) \to 0 \quad (r \to \infty).$$

Case 1.—Let $0 < s \leqslant \frac{1}{2}$.

Then, if $0 < z \leqslant 1$, $z^{\frac{1}{2}-s} \leqslant 1$,

and if $1 < z$, $z^{\frac{1}{2}-s} < z^{\frac{1}{2}}$.

Thus in either case $z^{\frac{1}{2}-s}J_m(z)$ is bounded, since $J_m(z)$ and $\sqrt{z}J_m(z)$ are both bounded.

Case 2.—Let $\frac{1}{2} < s \leqslant m+\frac{1}{2}$.

Then, if $0 < z \leqslant 1$, $z^{\frac{1}{2}-s} < z^{-m}$,

and if $1 < z$, $z^{\frac{1}{2}-s} < 1$.

Thus in either case $z^{\frac{1}{2}-s}J_m(z)$ is bounded, since $z^{-m}J_m(z)$ and $J_m(z)$ are both bounded.

Thus in all cases $z^{\frac{1}{2}-s}J_m(z)$ is bounded.

Now
$$A_r/C_r = \int_0^1 J_m(k_r t)\, t\, f(t)\, dt$$
$$= k_r^{s-\frac{1}{2}} \int_0^e \{(k_r t)^{\frac{1}{2}-s} J_m(k_r t)\}\, t^{\frac{1}{2}+s} f(t)\, dt$$
$$+ k^{-\frac{1}{2}} \int_e^1 \{(k_r t)^{\frac{1}{2}} J_m(k_r t)\}\, t^{\frac{1}{2}} f(t)\, dt.$$

Using in the second integral the asymptotic formula (viii), we see that it vanishes, by the theorem of Riemann-Lebesgue, when $r \to \infty$. Denoting by B the upper bound of $|(k_r t)^{\frac{1}{2}-s} J_m(k_r t)|$, the modulus of the first integral does not exceed

$$B \int_0^e |t^{\frac{1}{2}+s} f(t)|\, dt,$$

which is as small as we please, if e be chosen small enough. Thus this term determines the order $o(k_r^{s-\frac{1}{2}})$ of A_r/C_r.

Hence, since $C_r = O(k_r)$, and $k_r^{\frac{1}{2}} J_m(k_r z)$ is bounded,

$$k_r^{-s} A_r J_m(k_r z) \to 0.$$

This proves the theorem, since k_r is of order r.

6. The preceding theorem is important in connexion with the Cesàro summation of Bessel series, in virtue of the following theorem respecting the Cesàro convergence of Restricted Fourier Series :—

Theorem.—*The convergence (Ck) of a Restricted Fourier Series at any point, internal to the interval* $(-\pi < x < \pi)$, *depends only on the nature of the associated function in the neighbourhood of the point, provided only the typical term tends to zero when divided by the k-th power of the integer expressing its place in the series.**

For the proof of this theorem I refer to a forthcoming paper to be published elsewhere, as I do not propose to extend the length of the present communication by entering into a discussion of Cesàro convergence. It seems, however, relevant to the matter in hand to state and prove the above theorem respecting the convergence to zero of the typical Bessel term.

7. We proceed now to the discussion of the convergence of a Bessel series satisfying (6).

We shall require the following simple lemma :—

Lemma.—*If* dg/dt *and* dh/dt *exist and are bounded functions of* t, *and*

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* The importance of this result in the theory of the Cesàro convergence of series in general cannot be overrated. Thus, in the case of Legendre series, it at once leads to the theorem that *the Legendre series of* $f(x)$ *converges* (Ck), *where* $k \geqslant \frac{1}{2}$, *whenever the Fourier series of* $f(\cos\theta)$, *where* $x = \cos\theta$, *converges* (Ck), *where* $k > 0$.

In the case of Bessel series, a similar statement is true, as follows from the argument in the text, whenever $xf(x)$ is absolutely integrable. More generally, *if* $x^s f(x)$ *is absolutely integrable, the Bessel series of* $f(x)$ *converges* $(C, s+\frac{1}{2})$, *where* $0 \leqslant s < \frac{1}{2}$, *whenever the Fourier series of* $f(x)$ *converges* (Ck), *where* $k > 0$. The corresponding statements when $s \geqslant \frac{1}{2}$ may be left to the reader who has mastered the various papers concerned.

The difference between the Legendre and Bessel series will be easily understood; the convergence of the typical Legendre term to zero is secured by $(1-x^2)^{-\frac{1}{4}} f(x)$ being absolutely integrable, while, if this condition be not imposed, the typical Legendre term only tends to zero when divided by $\sqrt{n}$.

the repeated integral series

$$\Sigma \int g(t)\,dt \int h(t)\,u_r(t)\,dt$$

*converges uniformly to a repeated integral**

$$\Phi(t) \equiv \int g(t)\,dt \int h(t) f(t)\,dt,$$

then the second integrated series of $\Sigma u_r(t)$ *converges uniformly to the repeated integral of* $f(t)$, *i.e.*

$$\Sigma \int dt \int u_r(t)\,dt = \int dt \int f(t)\,dt.$$

The converse is also true.

Let us write

$$s_r = \sum_1^r \int u_r(t)\,dt, \qquad \sigma_r = \sum_1^r \int h(t)\,u_r(t)\,dt, \qquad \eta(t) = 1/h(t),$$

$$S_r = \sum_1^r \int dt \int u_r(t)\,dt, \quad \Sigma_r = \sum_1^r \int g(t)\,dt \int h(t)\,u_r(t)\,dt, \quad \gamma(t) = 1/g(t).$$

Then
$$g(t)\,\sigma_r(t) = \frac{d}{dt}\,\Sigma_r(t)$$

and therefore

$$s_r = \int \eta(t)\,\frac{d}{dt}\,\sigma_r(t)\,dt = \eta(t)\,\sigma_r(t) - \int \eta'(t)\,\gamma(t)\,\frac{d}{dt}\,\Sigma_r(t)\,dt$$

$$= \eta(t)\,\sigma_r(t) - \eta'(t)\,\gamma(t)\,\Sigma_r(t) + \int \frac{d}{dt}\,\{\eta'(t)\,\gamma(t)\}\,\Sigma_r(t)\,dt.$$

Hence, integrating,

$$S_r = \eta(t)\,\gamma(t)\,\Sigma_r(t) - \int \frac{d}{dt}\,\{\eta(t)\,\gamma(t)\}\,\Sigma_r(t)\,dt - \int \eta'(t)\,\gamma(t)\,\Sigma_r(t)\,dt$$

$$+ \int dt \int \frac{d}{dt}\,\{\eta'(t)\,\gamma(t)\}\,\Sigma_r(t)\,dt.$$

* Which can, of course, be expressed in the form given to $\Phi(t)$ since, $g'(t)$ and $h'(t)$ being bounded, $g(t)$ and $h(t)$ are integrals, and the hypothesis might be written

$$\Sigma \int dG \int u_r(t)\,dH = \int dG \int f(t)\,dH,$$

where G and H are the integrals of g and h.

Similarly,

$$\int dt \int f(t)\, dt = \eta(t)\, \gamma(t)\, \Phi(t) - \int \frac{d}{dt} \{\eta(t)\, \gamma(t)\} \Phi(t)\, dt - \int \eta(t)\, \gamma(t)\, \Phi(t)\, dt$$
$$+ \int dt \int \frac{d}{dt} \{\eta'(t)\, \gamma(t)\} \Phi(t)\, dt.$$

But since the series whose partial summation is Σ_r converges uniformly to $\Phi(t)$, the right-hand side of the last equation is the sum of the series whose partial summation is the right-hand side of the preceding one. Hence the required result at once follows.

The converse only requires the change of g into γ, and h into η, &c.

8. We next require the correlative in our theory of Lerch's theorem, that is, the theorem that *if two functions have the same Bessel series they only differ by a null-function, and, if continuous, are identical.*

For this purpose we need a subsidiary result, the proof of which is best obtained by using the complex variable.

Let us write [using $f(t)$ and $F(t)$ now without any reference to our former use of the symbol $f(t)$, but merely in order to maintain the notation in Lindelöf, *Calcul des Résidus*, pp. 20, 21]

$$f(t) = \frac{tJ'_m(t)}{J_m(t)} + H,$$

and therefore, using (iv),

$$f'(t) = \frac{(m^2 - t^2)\, J_m^2 - t^2 J_m'^2}{tJ_m^2}.$$

The zeros of $f(t)$ are then, by (2), our values $k_1, k_2, \ldots, k_r, \ldots$, taken both positively and negatively, none being zero; and its poles are the zeros of $J_m(t)$, excluding the origin, these necessarily distinct from the former.

Now put

$$F(t) = -\frac{J_m(tx)}{tJ_m(t)\, f'(t)} = \frac{J_m(tx)\, J_m(t)}{t^2} \Big/ \frac{1}{t^2} \{(t^2 - m^2)\, J_m^2 + t^2 J_m'^2\},$$

and integrate round a contour C, excluding those, say r, of the poles of $f(t)$ that lie within the contour by small circles K, the origin by a small circle O, and the roots of

$$(m^2 - t^2)\, J_m^2 - t^2 J_m'^2 = 0,$$

none of which coincide with k's, by small circles L. Then we get, by the

usual rule,

$$\frac{1}{2\pi i}\int_{C+K+O+L}\frac{F(t)f'(t)}{f(t)}\,dt = 2\{F(k_1)+F(k_2)+\ldots+F(k_r)\}.$$

The integrand here may be written

$$-\frac{J_m(tx)}{tJ'_m(t)+HJ_m}.$$

As the excluded poles of $f(t)$ are not poles of this function, we may now erase the corresponding circles K, and also the circles L, for the same reason, as $f(t)$ and $f'(t)$ do not vanish together.

Hence we have

$$-\frac{1}{2\pi i}\int_C \frac{J_m(tx)\,dt}{t\{HJ_m(t)+tJ'_m(t)\}}+\frac{1}{2\pi i}\int_O \frac{J_m(tx)\,dt}{t\{HJ_m(t)+tJ'_m(t)\}} = 2\sum_{n=1}^{r} F(k_n).$$

The second integral gives us then a residue

$$-\operatorname*{Lt}_{t=0}\frac{J_m(tx)}{HJ_m(t)+tJ'_m(t)} = -\frac{x^m}{H+m}.$$

Now the value of the first integral tends to zero when the contour C tends to infinity, as will be found if we use the known asymptotic expressions for J_m and J'_m* for complex values of the variable. The integrand then becomes of the order $1/(\sqrt{x}\,t^2)$. Hence, finally,

$$2\sum_{r=1}^{\infty} F(k_r) = -\frac{x^m}{m+H},$$

which is the result required and may be written in the form

$$x^m = (m+H)\,\Sigma C_r k_r^{-2} J_m(k_r)\,J_m(k_r x), \tag{7}$$

C_r being given by the equation (5).

9. We can now prove the following theorem, from which our extension of Lerch's theorem evidently follows immediately:—

THEOREM.—*If the coefficients of the Bessel series of a function $g(x)$ are all zero, $g(x)$ is a null-function.*

For we have proved (§ 8) by the complex variable that

$$\frac{x^m}{m+H} = \Sigma k_r^{-2}\,C_r J_m(k_r x)\,J_m(k_r). \tag{7}$$

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* For instance, formula (4) in Hobson's paper, *loc. cit.*, p. 364.

This series evidently converges uniformly in the closed interval $0 \leqslant x \leqslant B$. Hence, putting $x = \sqrt{z}$, we may multiply both sides by $z^{\frac{1}{2}m}$ and integrate term-by-term, from 0 to z. We thus get, by (vi),

$$z^{m+1}/(m+H)(m+1) = 2z^{\frac{1}{2}(m+1)} \Sigma k_r^{-3} C_r J_{m+1}(k_r\sqrt{z})\, J_m(k_r),$$

that is, $$z^{\frac{1}{2}(m+1)}/(m+H)(m+1) = 2\Sigma k_r^{-3} C_r J_{m+1}(k_r\sqrt{z})\, J_m(k_r).$$

Now multiply by $z^{-\frac{1}{2}(m+1)}$ and integrate between 0 and z. We get, by (vi),

$$\frac{z}{(m+H)(m+1)} = 4\Sigma \left\{ \frac{1}{2^m m!} - x^{-\frac{1}{2}m} J_m(k_r\sqrt{z}) \right\} J_m(k_r)\, C_r k_r^{-4},$$

or
$$\frac{z^{\frac{1}{2}m+1}}{(m+H)(m+1)} = 4\Sigma C_r k_r^{-4} J_m(k_r)\, z^{\frac{1}{2}m}/2^m m! - 4\Sigma C_r k_r^{-4} J_m(k_r)\, J_m(k_r\sqrt{z}). \quad (8)$$

This process may be repeated any number of times and we get similar equations for $z^{\frac{1}{2}m+p}$ for all integral values of p. From (7) we have, if $g(\sqrt{z})$ is an absolutely integrable function,

$$\int z^{\frac{1}{2}m}\, g(\sqrt{z})\, dz = (m+H)\, \Sigma C_r k_r^{-2} J_m(k_r) \int_0^1 J_m(k_r\sqrt{z})\, g(\sqrt{z})\, dz = 0,$$

under the supposition that the coefficients of the Bessel series of $g(\sqrt{z})$ are all zero.

Hence, also from (8), in the same circumstances,

$$\int_0^1 z^{\frac{1}{2}m+1}\, g(\sqrt{z})\, dz = 0,$$

and, from the similar equations with any integer p in place of 1,

$$\int_0^1 z^{\frac{1}{2}m+p}\, g(\sqrt{z})\, dz = 0.$$

Hence, by the extended theorem of Lerch, $z^{\frac{1}{2}m}\, g(\sqrt{z})$ is a null function, and therefore so is also $g(\sqrt{z})$. Moreover, if $g(\sqrt{z})$ is continuous in the open interval $(0 < x)$ it is necessarily zero for $0 < x$.

This proves the theorem.

10. Now if $f_1(t)$ and $f_2(t)$ have the same Bessel series, and we put

$$g(\sqrt{t}) = f_1(t) - f_2(t),$$

it follows, from the formal expression (4) for the coefficients of a Bessel series, that the coefficients of the Bessel series of $g(\sqrt{t})$ are all zero.

Hence by the preceding theorem $g(\sqrt{t})$ is a null function. Thus :—

THEOREM (*Lerch's Theorem for Bessel functions*).—*If two functions have the same Bessel series they only differ by a null-function, and, if they are continuous, they are identical.*

COR.—*If a series of Bessel functions converges uniformly, it is the Bessel series of a continuous function, and this continuous function is determined uniquely, and is the sum-function.*

For let
$$g(\sqrt{t}) = \Sigma A_r J_m(k_r \sqrt{t}), \tag{1}$$
where the convergence is uniform.

Let us multiply (1) term-by-term by $\frac{1}{2}C_r J_m(k_r \sqrt{t})$, and integrate from 0 to 1. Since (1) converges uniformly, this is allowable. Using the formulæ (5) and (iii), we then have, as the result of integration,

$$\tfrac{1}{2}C_r \int_0^1 g(\sqrt{t})\, J_m(k_r \sqrt{t})\, dt = A_r,$$

for all values of r. This shews that (1) is the Bessel series of $g(\sqrt{t})$.

But $g(\sqrt{t})$ is the sum of a uniformly convergent series of continuous functions, and is therefore continuous. Thus, by the preceding theorem, if the series (1) is also the Bessel series of $f(t)$ we see, comparing the two expressions obtained for C_r, that $f(\sqrt{t})$ and $g(\sqrt{t})$ can only differ by a null-function, and that, if $f(\sqrt{t})$ is continuous, it is identical with $g(\sqrt{t})$.

This proves the corollary.

11. We have next to prove that the second integrated series converges uniformly to a repeated integral. We deduce this from the following theorem :—

THEOREM.—*If the series* $\Sigma A_r J_m(k_r z)$, (1)

satisfying the condition $k_r^{-\frac{1}{2}} A_r \to 0 \quad (r \to \infty)$,

is the Bessel series of a function $f(z)$, *the series*

$$\Sigma V_r = \Sigma A_r \int_z^1 z^{-(m+1)} \left\{ \int_0^z t^{\frac{1}{2}m} J_m(k_r \sqrt{t})\, dt \right\} dz$$

converges uniformly to a function $\psi(z)$ *which is the integral of an integral, and has the form*

$$\psi(z) = \int_z^1 z^{-(m+1)} \left\{ \int_0^z t^{\frac{1}{2}m} f(\sqrt{t})\, dt \right\} dz.$$

By the formula (vi) we have at once

$$\int_0^z t^{\frac{1}{2}m} J_m\{k_r\sqrt{t}\}\,dt = \frac{2}{k_r} z^{\frac{1}{2}(m+1)} J_{m+1}(k_r\sqrt{z}). \tag{9}$$

Therefore, multiplying both sides by z^{-m-1}, and integrating

$$\begin{aligned}\int_z^1 z^{-m-1}dz \int_0^z t^{\frac{1}{2}m} J_m(k_r\sqrt{t})\,dt &= \frac{2}{k_r}\int_z^1 z^{-\frac{1}{2}(m+1)} J_{m+1}(k_r\sqrt{z})\,dz \\ &= \frac{2}{k_r}\int_{k_r^2 z}^{k_r^2} k_r^{m-1} u^{\frac{1}{2}(m+1)} J_{m+1}(\sqrt{u})\,du \\ &= -\frac{4}{k_r}\int_{k_r^2 z}^{k_r^2} k_r^{m-1}\frac{d}{du}\{u^{-\frac{1}{2}m} J_m(\sqrt{u})\}\,du \\ &= \frac{4}{k_r^2} z^{-\frac{1}{2}m} J_m(k_r\sqrt{z}) - \frac{4}{k_r^2} J_m(k_r). \end{aligned} \tag{10}$$

We notice particularly here that the second term of the right-hand side is got from the first by putting $z = 1$.

Now J_m is a bounded function, and by hypothesis $k_r^{-\frac{1}{2}} A_r \to 0$, and is therefore bounded. Thus the result just obtained shews that *the series* ΣV_r *is a uniformly convergent series.*

12. Now
$$A_r = \tfrac{1}{2} C_r \int_0^1 f(\sqrt{t})\, J_m(k_r\sqrt{t})\,dt. \tag{4}$$

Let us write
$$\phi(t) = \int_0^t u^{\frac{1}{2}m} f(\sqrt{u})\,du,$$

$$\psi(t) = \int_z^1 \phi(u)\, u^{-(m+1)}\,du;$$

then

$$\begin{aligned}\int_0^1 f(\sqrt{t})\, J_m(k_r\sqrt{t})\,dt &= \int_0^1 \phi'(t)\, t^{-\frac{1}{2}m} J_m(k_r\sqrt{t})\,dt \\ &= \Big[\phi(t)\, t^{-\frac{1}{2}m} J_m(k_r\sqrt{t})\Big]_0^1 + \tfrac{1}{2}\int_0^1 k_r\phi(t)\, t^{-\frac{1}{2}(m+1)} J_{m+1}(k_r\sqrt{t})\,dt \\ &= \phi(1)\, J_m(k_r) - \tfrac{1}{2}\int_0^1 k_r\psi'(t)\, t^{\frac{1}{2}(m+1)} J_{m+1}(k_r\sqrt{t})\,dt \\ &= \phi(1)\, J_m(k_r) + \tfrac{1}{4}\int_0^1 k_r^2\psi(t)\, t^{\frac{1}{2}m} J_m(k_r\sqrt{t})\,dt. \end{aligned}$$

Thus
$$A_r = \tfrac{1}{2} C_r \Big\{\phi(1)\, J_m(k_r) + \tfrac{1}{4}\int_0^1 k_r^2\psi(t)\, t^{\frac{1}{2}m} J_m(k_r\sqrt{t})\,dt\Big\}. \tag{11}$$

13. By (10) and (11) we get V_r expressed as the sum of four product terms, of which two are

$$2C_r \frac{\phi(1)\, J_m(k_r)}{k_r^2}\, z^{-\frac{1}{2}m} J_m(k_r\sqrt{z}) \tag{12}$$

and

$$\tfrac{1}{2}C_r z^{-\frac{1}{2}m} J_m(k_r\sqrt{z}) \int_0^1 \psi(t)\, t^{\frac{1}{2}m} J_m(k_r\sqrt{t})\, dt, \tag{13}$$

and the other two are got from these by putting $z = 1$. The series (13), when multiplied by $z^{\frac{1}{2}m}$, becomes the typical term of the Bessel series $\Sigma P_r J_m(k_r\sqrt{t})$ of the function $z^{\frac{1}{2}m}\psi(z)$.

But, as we have already seen (§ 8),

$$\frac{x^m}{m+H} = \Sigma k_r^{-2} C_r J_m(k_r x)\, J_m(k_r).$$

Therefore, putting $x = \sqrt{z}$, we see that (12) is the typical term of a series which converges to

$$4\phi(1)/(m+H),$$

and therefore, combined with the term got from this by putting $z = 1$, gives, after summation, the result zero. Thus, multiplying by $z^{\frac{1}{2}m}$, we have

$$z^{\frac{1}{2}m}\Sigma V_r = \{\Sigma P_r J_m(k_r\sqrt{t}) - P_r J_m(k_r)\}. \tag{14}$$

But, by (10) and (11),

$$P_r J_m(k_r) = \{A_r - \tfrac{1}{2}C_r\,\phi(1)\, J_m(k_r)\}\,\frac{4}{k_r^2} J_m(k_r), \tag{15}$$

and, since $k_r^{-\frac{1}{2}} A_r \to 0$ and $k_r^{-1} C_r$ and $k_r^{\frac{1}{2}} J_m$ are bounded, it follows that the series $\Sigma P_r J_m(k_r)$ converges uniformly. Hence, by (14), since, as we have seen, ΣV_r converges uniformly, the Bessel series $\Sigma P_r J_m(k_r\sqrt{t})$ converges uniformly.

By the Corollary of § 10, it follows that this series is the Bessel series of its sum function, and that this sum function is $z^{\frac{1}{2}m}\psi(z)$, since this latter function is continuous, and $\Sigma P_r J_m(k_r\sqrt{t})$ is its Bessel series.

But if we put $z = 1$, this sum-function is zero; therefore

$$\Sigma P_r J_m(k_r) = 0,$$

and we get, by (14),

$$z^{\frac{1}{2}m}\Sigma V_r = z^{\frac{1}{2}m}\psi(z),$$

or

$$\Sigma V_r = \psi(z),$$

which, as we have already proved that this series converges uniformly, completes the proof of the theorem.

14. From the preceding theorem, together with the Lemma of § 7, we have immediately the following

THEOREM.—*The second integrated series of the Bessel series*

$$\Sigma A_r J_m(k_r z)$$

of $f(z)$ *converges uniformly to* $\int dx \int f(z)\,dx$ *in* $(0 < z < 1)$.

We have also the following special corollary used later on :—

$$\Sigma \int dz \int \sqrt{z}\, A_r J_m(k_r z)\, dz = \int dz \int \sqrt{z}\, f(z)\, dz. \tag{16}$$

15. We now come to the fundamental theorem which enables us to treat Bessel series like Fourier series.

THEOREM.—*The Bessel series*

$$\sum_{r=0}^{\infty} A_r J_m(k_r z), \tag{1}$$

where

$$A_r = \tfrac{1}{2} C_r \int_0^1 f(\sqrt{t})\, J_m(k_r \sqrt{t})\, dt, \tag{4}$$

behaves in respect of convergence, divergence, oscillation, uniform or otherwise, precisely like the Fourier series of $f(z)$, *provided*

(1) *the point or points considered are internal to the completely open interval* $0 < x < 1$;

(2) *the individual terms of the series* (1) *tend to zero, that is*

$$k_r^{-\frac{1}{2}} A_r \to 0 \quad (r \to \infty).$$

To prove this we use the asymptotic expression (viii). We may write this in the form

$$A_r J_m(k_r z) = z^{-\frac{1}{2}} \{u_r + z^{-1} v_r + z^{-2} w_r\} \ldots,$$

where we use the following abbreviations :—

$$\psi_r = -k_r z + \tfrac{1}{4}(2m+1)\pi, \tag{17}$$

$$u_r = \frac{A_r}{\sqrt{k_r}} \sqrt{\frac{2}{\pi}} \cos \psi_r, \tag{18}$$

$$v_r = \frac{A_r}{\sqrt{k_r}} \sqrt{\frac{2}{\pi}}\, \tfrac{1}{2}(m^2 - \tfrac{1}{4})\, k_r^{-1} \sin \psi_r = \tfrac{1}{2}(m^2 - \tfrac{1}{4}) \int u_r dz, \tag{19}$$

$$w_r = -\frac{A_r}{\sqrt{k_r}} \sqrt{\frac{2}{\pi}}\, \frac{1}{2!\ 2^2} (m^2 - \tfrac{1}{4})(m^2 - \tfrac{9}{4})\, k_r^{-2} \left\{ \cos \psi + \frac{\theta(m^2-4)}{6 k_r z} \right\}, \tag{20}$$

where $|\theta| \leqslant 1$.

Thus writing for brevity, symbolically,

$$S_1 = \sum_{r=0}^{\infty} u_r, \qquad S_2 = \sum_{r=0}^{\infty} v_r, \qquad S_3 = \sum_{r=0}^{\infty} w_r,$$

we may write $$\sum_{r=0}^{\infty} A_r J_m(k_r z) = z^{-\frac{1}{2}} \{S_1 + z^{-1} S_2 + z^{-2} S_3\}, \tag{21}$$

this relation being understood not to imply convergence of the series involved, but merely identity, when we replace the infinite series by the corresponding n-th partial summations.

It is, however, at once evident, from the form of w_r, that S_3 is a uniformly convergent series, since $k_r^{-\frac{1}{2}} A_r$ is bounded.*

16. Since, then, we have seen that the second integrated series of the Bessel series $\Sigma A_r J_m(k_r z)$ converges uniformly to the integral of an integral, and S_3 converges uniformly, it follows that, *in the closed interval* $(e \leqslant z \leqslant 1)$, *where* $0 < e$,

$$\Sigma A_r J_m(k_r z) - z^{-\frac{3}{2}} S_3 = z^{-\frac{1}{2}} S_1 + z^{-\frac{3}{2}} S_2,$$

converges uniformly to the integral of an integral, when twice integrated term by term.

17. But, from the form of v_r, S_2, when *once* integrated, converges, like S_3, uniformly, under the same condition as to A_r. Therefore in the closed interval $(e, 1)$, $z^{-\frac{1}{2}} S_1$, integrated *twice*, converges uniformly to the difference between an integral of an integral and the integral of a continuous function, that is to an integral; the same is therefore true of S_1 itself, when twice integrated.

Hence S_2—whose individual term is a multiple of the integral of that of S_1—when *once* integrated, converges uniformly to an integral. The same is therefore true of $z^{-\frac{3}{2}} S_2$ in the closed interval $(e, 1)$.

It follows, since $$z^{-\frac{3}{2}} S_2$$

and $$z^{-\frac{1}{2}} S_1 + z^{-\frac{3}{2}} S_2,$$

when *twice* integrated, each converge to the integral of an integral in $(e, 1)$, that the same is true of $z^{-\frac{1}{2}} S_1$; and therefore also S_1 *when integrated twice converges uniformly in* $(e, 1)$ *to the integral of an integral.*

We shall see that we can prove more than this, and that S_1 when once integrated converges uniformly to an integral.

* It suffices here that $A_r k_r^{-s}$ is bounded, with $s < \frac{3}{2}$.

18. At this point, for reasons which will appear subsequently, we shall make the assumption that m is rational with μ for denominator. It must not, however, be supposed that there is any such restriction involved in the theorem itself.

By (3) we may then write

$$k_r = \frac{\pi}{4\mu}(R+\beta_r), \tag{22}$$

where R is an integer which increases by 4μ each time r increases by 1, and

$$\beta_r = \beta R^{-1} + O(R^{-2}). \tag{23}$$

Writing ϵ_1 and ϵ_2 for the sine and cosine of $\frac{1}{4}(2m+1)\pi$, we then have, by (17), putting

$$y = \pi z/4\mu, \tag{24}$$

$$\begin{aligned}\cos\psi_r &= \epsilon_1 \cos k_r z + \epsilon_2 \sin k_r z \\ &= (\epsilon_1 \cos Ry + \epsilon_2 \sin Ry)\cos\beta_r y + (\epsilon_2 \cos Ry - \epsilon_1 \sin Ry)\sin\beta_r y.\end{aligned}$$

Thus, writing

$$\tau_r = \sqrt{\frac{2}{\pi k_r}} A_r(\epsilon_1 \cos Ry + \epsilon_2 \sin Ry), \tag{25}$$

$$\tau_{1,r} = \sqrt{\frac{2}{\pi k_r}} A_r(\epsilon_2 \cos Ry - \epsilon_1 \sin Ry), \tag{26}$$

$$T = \sum_{r=0}^{\infty} \tau_r, \tag{27}$$

$$T_1 = \sum_{r=1}^{\infty} R^{-1}\tau_{1,r} = -\Sigma\int \tau_r\, dy, \tag{28}$$

we get

$$\begin{aligned}S_1 = \sum_{r=0}^{\infty} u_r &= \sum_{r=0}^{\infty} \sqrt{\frac{2}{\pi k_r}} A_r \cos\psi_r = \sum_{r=0}^{\infty} (\tau_r \cos\beta_r y + \tau_{1,r}\sin\beta_r y) \\ &= T + \beta y T_1 + T_2,\end{aligned} \tag{29}$$

where T_2 is a uniformly convergent series whose sum, as we shall next show, is an integral.

19. Indeed, since τ_r and $\tau_{1,r}$ are bounded, and

$$\frac{d}{dy}\tau_r = R\tau_{1,r},$$

$$\frac{d}{dy}\tau_{1,r} = -R\tau_r,$$

we have $$\frac{d}{dy}(u_r-\tau_r-\beta_r\tau_{1,r}\,y/R+\tfrac{1}{2}\beta^2\tau_r y^2/R^2)$$

$$= -\frac{d}{dy}\{\tau_r(1-\tfrac{1}{2}\beta^2y^2/R^2-\cos\beta_r y)+\tau_{1,r}(\beta_r y/R-\sin\beta_r y)\}$$

$$= O(R^{-2}).$$

Thus $$\Sigma\frac{d}{dy}(u_r-\tau_r-\beta_r\tau_{1,r}\,y/R+\tfrac{1}{2}\beta^2\tau_r y^2/R^2),$$

is a uniformly convergent series. Integrating this term-by-term, we see that

$$S-T-\beta_r yT_1+\tfrac{1}{2}\beta^2y^2\Sigma\tau_r/R^2 = T_2+\tfrac{1}{2}\beta^2y^2\Sigma\tau_r/R^2$$

converges uniformly to an integral.

Since $\Sigma\tau_r/R^2$ is the first integrated series of T_1, we shall see by the next article that this proves the statement made at the end of the last article.

20. Now by (28) and (26) we see that T_1 is the Fourier series of a function of y whose square is absolutely integrable, since the sum of the squares of its coefficients is $\frac{2}{\pi}\sum_{r=0}^{\infty}(k_r^{-\frac{1}{2}}A_rR^{-1})^2$, which is convergent since

$$k_r^{-\frac{1}{2}}A_r\to 0.*$$

Therefore the first integrated series of T_1 converges uniformly, in the present circumstances, to an integral.

21. Integrating twice term-by-term the series

$$S_1-\beta yT_1-T_2,$$

we get therefore, by §§ 17, 20, 18 and 19, a uniformly convergent series whose sum is an integral. That is, since, by (29), each term of this series is equal to the corresponding term of T, *the second integrated series of T converges uniformly to an integral.*

But T is, by (27) and (25), a trigonometrical series whose coefficients tend to zero, since $k_r^{-\frac{1}{2}}A_r\to 0$ $(r\to\infty)$. Thus T is an R. F. series in the interval $(\epsilon\leqslant z\leqslant 1)$, that is in $(\pi\epsilon/4\mu\leqslant y\leqslant\pi/4\mu)$. Since ϵ is as small as we please, this proves that *T is an R. F. series in the interval $(0\leqslant y\leqslant\pi/4\mu)$, m being rational with μ for denominator.*

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* This would still be the case of course if $A_rk_r^{-s}$ were bounded, where $s<\frac{3}{2}$.

Hence in this interval T_1, the first integrated series of T_1, converges to an integral, the convergence being uniform in every closed interval inside the interval of restriction.

Thus we have proved that

$$S_1 = T+\beta T_1 y+T_2, \tag{29}$$

where T is an R. F. series in $(0 < y \leqslant \pi/4m)$, and T_1 is the integrated series of T, and is accordingly a Fourier series, which in $(0 < y \leqslant \pi/4m)$ converges uniformly to an integral, while T_2 is a uniformly convergent series, whose sum is an integral.

Also we had

$$\sum_{r=0}^{\infty} A_r J_m(k_r z) = z^{-\frac{1}{2}}(S_1+z^{-1}S_2+z^{-2}S_3), \tag{21}$$

where a constant multiple of S_2 is the integrated series of S_1, and therefore S_2 converges uniformly to an integral in every closed interval in $(0 < y \leqslant \pi/4m)$ or $(0 < z \leqslant 1)$, and S_3 is a uniformly convergent series whose sum, as we shall now prove, is an integral.

22. In fact, by (19) and (20),

$$w_r = (m^2-\tfrac{9}{4})\frac{1}{2!}\int v_r dz+A_r Q_m(k_r z), \tag{22}$$

where, by the formula (viii),

$$Q_m(x) = \sqrt{\left(\frac{\pi x}{2}\right)} J_m(x)-\cos\psi-\frac{m^2-\frac{1}{4}}{2x}\sin\psi+\frac{(m^2-\frac{1}{4})(m^2-\frac{9}{4})}{2!\,(2x)^2}\cos\psi. \tag{23}$$

It is therefore only necessary to shew that $\sum_{r=0}^{\infty} A_r k_r^{-\frac{1}{2}} Q_m(k_r z)$ converges uniformly to an integral, since the first term on the right in (22), when summed, converges uniformly to a numerical multiple of the integral of the sum-function of S_2, *i.e.* to the integral of an integral.

Now, differentiating (23), we see that $\Sigma A_r k_r^{-\frac{1}{2}} \frac{d}{dx} Q_m(k_r z)$ is a uniformly convergent series*; therefore, integrating term-by-term, $\Sigma A_r k_r^{-\frac{1}{2}} Q_m(k_r z)$ converges uniformly to an integral.

Hence, as stated, the sum of the series S_3 is an integral.

* It is only necessary that $A_r k_r^{-s}$ should be bounded, with $s < \frac{5}{2}$.

23. Thus, finally,

$$\Sigma A_r J_m(k_r z) = z^{-\frac{1}{2}}(S_1 + z^{-1} S_2 + z^{-2} S_3), \tag{21}$$

where S_2 and S_3 are uniformly convergent series, whose sums are integrals, and

$$S_1 = T + \beta T_1 y + T_2, \tag{29}$$

where T_1 and T_2 are uniformly convergent series whose sums are integrals, while T is an R. F. series in $(0 < y \leqslant \pi/4\mu)$, where $y = z/4\mu$.

It follows that $\Sigma A_r J_m(k_r z)$ converges, or diverges, according as this R. F. series does so. Moreover the relations (21) and (29) are identities when we replace each of the infinite series involved by the corresponding n-th partial summations, so that, even when the R. F. series and the Bessel series do not converge, their oscillation properties are identical, and in particular uniform convergence or oscillation, or bounded convergence or oscillation, of the one series entails the same property in the other. Also term-by-term integration properties of the one series may be immediately inferred for one of the series if they are known to hold for the other.

24. Thus we may write

$$\Sigma A_r J_m(k_r z) = z^{-\frac{1}{2}} \{T + \phi(z)\}, \tag{30}$$

where $\phi(z)$ is an integral.

Integrating twice after multiplying by $\sqrt{z}$, and denoting the function associated with T by $g(y)$,

$$\Sigma \int dz \int \sqrt{z}\, f(z)\, dz = \int dz \int g\left(\frac{\pi z}{4\mu}\right) dz + \int dz \int \phi(z)\, dz,$$

whence, differentiating twice at any internal point of the interval $(e, 1)$,

$$g\left(\frac{\pi z}{4\mu}\right) = \sqrt{z}\, f(z) - \phi(z),$$

or

$$g(y) = \sqrt{\left(\frac{4\mu y}{\pi}\right)} f\left(\frac{4\mu y}{\pi}\right) - \phi\left(\frac{4\mu y}{\pi}\right).$$

But the Fourier series of $\sqrt{z}\, f(z)$ converges or diverges at all points of $(e, 1)$ according as that of $f(z)$ does so. Thus our R. F. series converges or diverges according as the Fourier series of $f(z)$ does so.

Thus, finally, the Bessel series $\Sigma A_r J_m(k_r z)$, whose associated function is $f(z)$, converges, diverges, or oscillates, precisely like the Fourier series of $f(z)$.

This proves the theorem enunciated in § 15, when m is rational.

25.* The above was proved on the supposition that m was a rational number. This is not necessary, but was convenient, since it enabled us to use the existing mathematical machinery. We shall now shew that the result is the same, and the general trend of the reasoning also, when m is irrational.

In this case, writing $k = \frac{1}{4}(2m+1+4s)$,

$$\pi z = \pi - y,$$

$$\beta_r = \frac{\pi a(r)}{r} = \frac{\beta}{r} + O(r^{-2}), \tag{23a}$$

we have $$\psi_r = (k+r)\,y - (r+s)\,\pi - \beta_r z. \tag{17a}$$

Therefore, putting $$\tau_r = (-)^{r+s} A_r \sqrt{\left(\frac{2}{\pi k_r}\right)} \cos(k+r)y, \tag{25a}$$

$$\tau_{1,r} = (-)^{r+s} A_r \sqrt{\left(\frac{2}{\pi k_r}\right)} \sin(k+r)y, \tag{26a}$$

and using T, T_1, and S_1 in the same sense as before, we have, as in § 18,

$$S_1 = \sum_0^\infty u_r = \sum_0^\infty \sqrt{\left(\frac{2}{\pi k_r}\right)} A_r \cos\psi_r = T + \beta T_1 z + T_2. \tag{29}$$

The argument from this point is unaltered in character till we arrive at § 21 : for, though T_1 is no longer a Fourier series, it is the sum of two Fourier series, multiplied respectively by $\cos ky$ and $\sin ky$. Thus it is still true that the first integrated series of T_1 converges uniformly to an integral.

26. We thus see, as in § 21, that the second integrated series of T converges uniformly to the integral of an integral. But T is not any longer an R. F. series.

We proceed therefore to make a short digression on a non-harmonic trigonometrical series

$$\sum_{r=0}^\infty \{a_r \cos(k+r)\,x + b_r \sin(k+r)\,x\},$$

whose individual term tends to zero, while its second integrated series converges uniformly to the integral of an integral.

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* Modified January 29th, 1918.

27. Let the series

$$-\sum_{r=0}^{\infty}(r+k)^{-2}\{a_r\cos(r+k)x+b_r\sin(r+k)x\} \quad (0<k<1),$$

where

$$\left.\begin{aligned} a_n &\to 0,\\ b_n &\to 0,\end{aligned}\right\} \quad (n\to\infty),$$

converge uniformly, and in the interval $(\alpha \leqslant x \leqslant \beta)$ let the sum-function $G(x)$ be a repeated integral

$$\int dx\int f(x)\,dx.$$

Then

$$\tfrac{1}{2}\{G(x+t)+G(x-t)\} = -\sum_{r=0}^{\infty}\frac{\cos(r+k)t}{(r+k)^2}\{a_r\cos(r+k)x+b_r\sin(r+k)x\}$$

$$= -\sum_{r=0}^{\infty}(r+k)^{-2}A_r\cos(r+k)t, \qquad (31)$$

and, the convergence being uniform,

$$\int_0^{\infty}\tfrac{1}{2}\{G(x+t)+G(x-t)\}\frac{1-\cos t}{t^2}\,dt$$

$$= -\sum_{r=0}^{\infty}(r+k)^{-2}A_r\int_0^{\infty}\frac{1-\cos t}{t^2}\cos(k+r)t\,dt. \qquad (32)$$

Now

$$2(1-\cos t)\cos(k+r)t = 2\cos(k+r)t-\cos(k+r+1)t-\cos(k+r-1)t.$$

Therefore, since $$\int_0^{\infty}\frac{1-\cos qt}{t^2}dt = \frac{\pi}{2}|q|,$$

the integral under the sign of summation in (32) is zero if $r \geqslant 1$, and when $r=0$ has the value

$$\tfrac{1}{2}k^{-2}A_0\frac{\pi}{2}(2-2k) = \frac{\pi}{2}k^{-2}A_0(1-k).$$

Thus $$\frac{2}{\pi}\int_0^{\infty}\tfrac{1}{2}\{G(x+t)+G(x-t)\}\frac{1-\cos t}{t^2}\,dt = -k^{-2}A_0(1-k). \qquad (33)$$

Again, similarly,

$$\int_0^{\infty}\tfrac{1}{2}\{G(x+t)+G(x-t)\}\cos t\,\frac{1-\cos t}{t^2}\,dt$$

$$= -\sum_{r=0}^{\infty}\tfrac{1}{2}(r+k)^{-2}A_r\int_0^{\infty}2\cos t\cos(k+r)t\,\frac{1-\cos t}{t^2}\,dt. \qquad (34)$$

Now

$$2\cos(k+r)\,t\cos t-2\cos(k+r)\,t\cos^2 t$$

$$=\cos(k+r+1)\,t+\cos(k+r-1)\,t-\cos(k+r)\,t-\cos(k+r)\,t\cos 2t$$

$$=\cos(k+r+1)\,t+\cos(k+r-1)\,t-\cos(k+r)\,t-\tfrac{1}{2}\cos(k+r+2)\,t$$

$$-\tfrac{1}{2}\cos(k+r-2)\,t\,;$$

whence, as before, all the integrals on the right of (34) vanish except for $r=0$, $r=1$, and we get

$$\frac{2}{\pi}\int_0^\infty \{G(x+t)+G(x-t)\}\cos t\,\frac{1-\cos t}{t^2}\,dt=-k^{-2}A_0k-(k+1)^{-2}A_1(1-k). \tag{35}$$

For all values of $n>1$, we have then, in like manner,

$$\int_0^\infty \tfrac{1}{2}\{G(x+t)+G(x-t)\}\cos nt\,\frac{1-\cos t}{t^2}\,dt$$

$$=-\sum_{r=0}^{\infty}(r+k)^{-2}A_r\int_0^\infty \cos(r+k)\,t\cos nt\,\frac{1-\cos t}{t^2}\,dt. \tag{36}$$

Now since

$$2\cos(r+k)\,t\cos nt\,(1-\cos t)$$

$$=\cos(r+k+n)\,t+\cos(r+k-n)\,t-\cos(r+k)\,t\cos(n+1)\,t$$

$$-\cos(r+k)t\cos(n-1)\,t$$

$$=\cos(r+k+n)\,t+\cos(r+k-n)\,t-\tfrac{1}{2}\cos(r+k+n+1)\,t$$

$$-\tfrac{1}{2}\cos(r+k+n-1)\,t-\tfrac{1}{2}\cos(r+k-n+1)\,t-\tfrac{1}{2}\cos(r+k-n-1)\,t,$$

the integrals on the right of (36) all vanish except for $r=n-1$, when the value is $\frac{1}{4}\pi k$, and for $r=n$, when it is $\frac{1}{4}\pi(1-k)$.

Thus, for $n>1$,

$$\frac{2}{\pi}\int_0^\infty \{G(x+t)+G(x-t)\}\cos nt\,\frac{1-\cos t}{t^2}\,dt$$

$$=-(k+n-1)^{-2}A_{n-1}k-(k+n)^{-2}A_n(1-k),$$

or, differentiating twice,

$$\frac{2}{\pi}\frac{d^2}{dx^2}\int_0^\infty \{G(x+t)+G(x-t)\}\cos nt\,\frac{1-\cos t}{t^2}\,dt=kA_{n-1}+(1-k)A_n. \tag{37}$$

Summing, and taking account of (35) and (33), we get

$$\frac{2}{\pi}\frac{d^2}{dx^2}\int_0^\infty \{G(x+t)+G(x-t)\}\,\tfrac{1}{2}\operatorname{cosec}\tfrac{1}{2}t\sin(n+\tfrac{1}{2})\,t\,\frac{1-\cos t}{t^2}\,dt$$

$$= \sum_{r=0}^{n} A_r + kA_n. \quad (38)$$

28.* Writing

$$\operatorname{cosec}\tfrac{1}{2}\sin(n+\tfrac{1}{2})t = \sin nt\cot\tfrac{1}{2}t - \cos nt,$$

we have the left-hand side of (38) expressed as the sum of two terms, the second of which, being by (37) equal to $kA_{n-1}+(1-k)A_n$, tends to zero as $n\to\infty$. Thus we are left with the first term, which we write $4I_n/\pi$, where

$$I_n = \frac{d^2}{dx^2}\int_0^\infty \tfrac{1}{2}\{G(x+t)+G(x-t)\}\sin nt\cot\tfrac{1}{2}t\,\frac{1-\cos t}{t^2}\,dt$$

$$= \frac{d^2}{dx^2}\int_0^\infty \tfrac{1}{2}\{G(x+t)+G(x-t)\}\,t^{-2}\sin t\sin nt\,dt.$$

We now divide the interval $(0,\infty)$ up into $(0,2\pi)$, $(2\pi,4\pi)$, ..., and sum the corresponding integrals, using the formula

$$\sin 2k\pi\,.\,\tfrac{1}{2}\{G[x+(2p\pi+t)]+G[x-(2p\pi+t)]\}$$

$$= -\sum_{r=1}^{\infty}(r+k)^{-2}A_r\cos\{(r+k)(2p\pi+t)\}\sin 2k\pi$$

$$= \cos 2kp\pi\sin 2k\pi\,.\,\tfrac{1}{2}\{G(x+t)+G(x-t)\}$$

$$-\sin 2kp\pi\sin 2k\pi\,.\,\tfrac{1}{2}\{G(x+t)-G(x-t)\}$$

$$= \sin 2k(p+1)\pi\,.\,\tfrac{1}{2}\{G(x+t)+G(x-t)\}$$

$$-\sin 2kp\pi\,.\,\tfrac{1}{2}\{G[x+(2\pi-t)]+G[x-(2\pi-t)]\}.$$

Hence

$$I_n = \frac{d^2}{dx^2}\Big[\int_0^{2\pi}\tfrac{1}{2}\{G(x+t)+G(x-t)\}\,\chi_1(t)\sin t\sin nt\,dt$$

$$-\int_0^{2\pi}\tfrac{1}{2}\{G[x+(2\pi-t)]+G[x-(2\pi-t)]\}\,\chi_2(2\pi-t)\sin t\sin nt\,dt\Big]$$

$$= \frac{d^2}{dx^2}\int_0^{2\pi}\tfrac{1}{2}\{G(x+t)+G(x-t)\}\{\chi_1(t)-\chi_2(t)\}\sin t\sin nt\,dt,$$

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* Modified January 29th, 1918.

where $$\chi_1(t) = t^{-2} + \operatorname{cosec} 2k\pi \sum_{p=1}^{\infty} (2p\pi + t)^{-2} \sin 2k(p+1)\pi,$$

$$\chi_2(2\pi - t) = \operatorname{cosec} 2k\pi \sum_{p=1}^{\infty} (2p\pi + t)^{-2} \sin 2kp\pi.$$

Thus $$\chi_1(2\pi) = \chi_2(2\pi).$$

Hence if we define a function $\lambda(t)$ as equal to $\chi_1(t) - \chi_2(t)$ from e to 2π, we shall have $\lambda(2\pi) = 0$, and $\lambda(t)$ is the integral of an integral in this interval. If therefore we assign suitable values to $\lambda(t)$ in $(0, e)$, this function will be the integral of an integral in $(0, 2\pi)$, and have zero for its values at 0 and 2π. Then $\lambda(t) \sin t$ will be the integral of an integral in $(0, 2\pi)$, with a zero value and a zero differential coefficient at each of the end-points 0 and 2π.

By a theorem which I have recently proved,* we have therefore uniformly,

$$\frac{d^2}{dx^2} \int_0^{2\pi} \tfrac{1}{2} \{G(x+t) + G(x-t)\} \lambda(t) \sin t \sin nt\, dt \to 0 \quad (n \to \infty).$$

Hence we may replace I_n by

$$\frac{d^2}{dx^2} \int_0^{2\pi} \tfrac{1}{2} \{G(x+t) + G(x-t)\} \{\chi_1(t) - \chi_2(t) - \lambda(t)\} \sin t \sin nt\, dt$$

$$= \frac{d^2}{dx^2} \int_0^{e} \tfrac{1}{2} \{G(x+t) + G(x-t)\} \{\chi_1(t) - \chi_2(t) - \lambda(t)\} \sin t \sin nt\, dt$$

$$= \int_0^{e} \tfrac{1}{2} \{f(x+t) + f(x-t)\} \{\chi_1(t) - \chi_2(t) - \lambda(t)\} \sin t \sin nt\, dt,$$

since $G(t)$ is the integral of an integral in $(x-e, x+e)$ and $f(t) = d^2G/dt^2$.

Now $$\chi_1(t) - \chi_2(t) - \lambda(t) - t^{-2}$$

is a bounded function of t in $(0, e)$; therefore

$$\tfrac{1}{2} \{f(x+t) + f(x-t)\} \{\chi_1(t) - \chi_2(t) - \lambda(t) - t^{-2}\} \sin t$$

is a summable function in $(0, e)$. Hence when we multiply by $\sin nt$ and integrate from 0 to e, the integral tends uniformly to zero by the theorem of Riemann-Lebesgue. We may therefore replace I_n by

$$\int_0^{e} \tfrac{1}{2} \{f(x+t) + f(x-t)\} t^{-2} \sin t \sin nt\, dt.$$

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* "On Non-Harmonic Trigonometrical Series," 1917, *Roy. Soc. Proc.* (A), Vol. 95, pp. 237-249.

29. Hence we have

$$\frac{1}{\pi}\int_0^e \{f(x+t)+f(x-t)\}\sin nt\cot\tfrac{1}{2}t\,\frac{1-\cos t}{t^2}\,dt+\eta_n=\sum_{r=0}^{n}A_r,$$

where $\eta_n\to 0$ uniformly.

Let us now write $\dfrac{1-\cos t}{t^2}=\tfrac{1}{2}+t^2\dfrac{1-\frac{1}{2}t^2-\cos t}{t^4}$.

Then since $$t^2\cot\tfrac{1}{2}t\,\frac{1-\frac{1}{2}t^2-\cos t}{t^4}$$

is a bounded function in $(0, e)$, the part of our integral involving this factor in the integrand tends to zero by the theorem of Riemann-Lebesgue.

Thus, finally,

$$\frac{1}{2\pi}\int_0^e \{f(x+t)+f(x-t)\}\sin nt\cot\tfrac{1}{2}t\,dt+\eta_n = \sum_{r=0}^{n}\{a_r\cos(r+k)\,x+b_r\sin(r+k)\,x\}, \quad (39)$$

where, as $n\to\infty$, $\eta_n\to 0$ uniformly.

30. The expression here found for the partial summation of our non-harmonic trigonometrical series is, barring an additive term which tends to zero as $n\to\infty$, the same as the partial summation of the Fourier series of $f(x)$. Indeed in the paper quoted this form of the partial summation is easily obtained [see equations (6) and (9) of that paper].

We may accordingly write in the more familiar expression for the partial summation of the Fourier series, and get

$$\frac{1}{2\pi}\int_0^e \{f(x+t)+f(x-t)\}\sin(n+\tfrac{1}{2})\,t\,\operatorname{cosec}\tfrac{1}{2}t\,dt+\eta_n = \sum_{r=0}^{n}\{a_r\cos(r+k)\,x+b_r\cos(r+k)\,x\}, \quad (40)$$

where $\eta_n\to 0$ uniformly.

Thus we see that, at a point x where $\alpha<x<\beta$, the upper and lower functions of the non-harmonic trigonometrical series

$$\sum_{r=0}^{\infty}\{a_r\cos(r+k)\,x+b_r\sin(r+k)\,x\},$$

where (i) $a_r\to 0, \quad b_r\to 0,$

and (ii) *the sum of the second integrated series is in* (α, β) *the integral of*

the integral of a summable function $f(x)$, are the same as those of the Fourier series of $f(x)$, and the difference of corresponding partial summations tends uniformly to zero.

31. Using this result in place of the corresponding theorem in the theory of R. F. series, the theorem of § 15 follows as before, m being now not necessarily rational.

Thus the fundamental theorem of § 15 has been established in all its generality.

More accurately what we have proved may be stated in the following form :—

Theorem.—*The upper and lower functions of the Bessel series of $f(x)$, at a point x internal to the interval $0 < x < 1$, are equal to those of the Fourier series of $f(x)$, and the corresponding partial summations differ only by a quantity which tends uniformly to zero, provided the individual terms of the Bessel series tend to zero.*

32. We have immediately the following corollaries :—

Cor. 1.—*The first integrated series*

$$\sum_{r=0}^{\infty} \int A_r J_r(k_r z)\, dz$$

of a Bessel series whose individual terms tend to zero, that is

$$k^{-\frac{1}{2}} A_r \to 0 \quad (r \to \infty),$$

converges uniformly to an integral at every point and in every interval in $(0 < z < 1)$.

For, in equation (30), we have seen that the integrated series of T converges uniformly to an integral in any closed interval inside $(0<z<1)$. Hence, multiplying by $z^{-\frac{1}{2}}$ and integrating (30) term-by-term, which is allowable by reason of the uniform convergence, the required result appears immediately.

33. The following extension of the main theorem is important :—

Cor. 2.—*The theorem is still true for a series of Bessel functions of the form* (1), *whose individual terms tend to zero, that is for any series*

$$\Sigma A_r J_m(k_r z), \tag{1}$$

where

$$k_r^{-\frac{1}{2}} A_r \to 0 \quad (r \to \infty),$$

provided in some interval containing the point, or points, under consideration the sum of the second integrated series is the integral of an integral.

In fact the condition (4), which makes our series in § 15 a Bessel series, nowhere enters into the proof itself of the main theorem, except tacitly, in quoting the theorem that the second integrated series converges to a repeated integral.

Hence, as in Cor. 1, *the first integrated series of such a series as we here contemplate converges to an integral.*

The function associated with the series of Bessel functions will then be the differential coefficient of the sum-function of the integrated series, in the interval under consideration in which this latter function is an integral.

It is evidently sufficient for the fulfilment of the conditions of the above theorem, if

$$k_r^{-\frac{1}{2}} A_r \to 0 \quad (r \to \infty),$$

and the integrated series converges *uniformly* to an integral.

34. It is unnecessary to enumerate all the results for Bessel series and series of Bessel functions which can now be transferred wholesale from the theory of Fourier series to that of these series. These theorems include, beside sufficient conditions for convergence, uniform convergence or oscillation, bounded convergence or oscillation, and so forth, as well as for Cesàro convergence, all theorems involving term-by-term integration when multiplied by suitable functions, provided we remain in the interior of the completely open interval (0, 1).

As an example we may quote one of the simplest of those theorems which are an immediate consequence of the main theorem :—

The Bessel series of a function which, in an interval (α, β), *is a continuous function of bounded variation converges uniformly to that function in* (α, β); *in particular this is the case in any interval* (α, β) *in which the function is an integral.*

Hence we have the following important result :—

THEOREM.—*If* $f(x)$ *is, in a neighbourhood surrounding the point* x, *the integral of a function whose Fourier series converges at that point, e.g. if it is the integral of a continuous function of bounded variation in such a neighbourhood, then the derived series of the Bessel series of* $f(x)$ *will converge at the point in question to the value* $f'(x)$.

For let the series be $\Sigma A_r J_m(k_r x)$,

where $$A_r = C_r \int_0^1 t f(t) J_m(k_r t)\, dt,$$

$f(t)$ being an integral in (α, β).

Then, by the formula (vi), if $m \geqslant 1$,

$$k_r \int_0^x t^m J_{m-1}(k_r t)\, dt = x^m J_m(k_r x).$$

Thus the series $\Sigma k_r A_r J_{m-1}(k_r x)$

is a series of Bessel functions, which, when multiplied by x^m and integrated term-by-term, converges uniformly to an integral, namely to $x^m f(x)$.

Therefore the integrated series of this series converges uniformly to $f(x)$.

If therefore the condition $\sqrt{k_r} A_r \to 0 \quad (r \to \infty)$

be satisfied, we can apply our theorem, and the theorem as enunciated at once follows. It will be seen by the following theorem that this condition is satisfied, which proves the required result.

35. Now, if $f(x)$ is an integral, $f'(x)$ is summable, and therefore in any interval in which $f(x)$ is an integral, and which does not contain the origin, $\sqrt{x} f'(x) - m f(x)/\sqrt{x}$ is summable, and $x^{m+1} f(x) \to 0$ as $x \to 0$. Thus we see that the result required to complete the preceding proof is a special case of the following theorem.

Theorem. — *If*

(i) $x^{m+\frac{1}{2}}\, d/dx \{x^{-m} f(x)\}$ *is summable in* $(0, 1)$,

and (ii) $x^{m+1} f(x) \to 0 \quad (x \to 0)$,

then $$\int_0^1 J_m(k_r x)\, x f(x)\, dx = O(k_r^{-\frac{3}{2}}).$$

Let us write $g(x)$ for the summable function involved in the hypothesis (i).

Then, by the formula (vi),

$$k_r \int_\epsilon^1 J_m(k_r x)\, x f(x)\, dx$$

$$= \int_\epsilon^1 x^{-m} f(x)\, d \int_0^x k_r x^{m+1} J_m(k_r x)\, dx$$

$$= f(1)\, J_{m+1}(k_r) - f(\epsilon)\, J_{m+1}(k_r \epsilon) + \int_\epsilon^1 x^{m+1} J_{m+1}(k_r x)\, \frac{d}{dx} \{x^{-m} f(x)\}\, dx$$

$$= f(1)\, \frac{m+H}{k_r}\, J_m(k_r) - f(\epsilon)\, J_{m+1}(k_r \epsilon) + \int_\epsilon^1 g(x) x^{\frac{1}{2}} J_{m+1}(k_r x)\, dx.$$

Now $x^{\frac{1}{2}} J_{m+1}(k_r x)$ is bounded in (0, 1) and $g(x)$ is summable. Therefore, if we let $\epsilon \to 0$, the integral on the right will exist up to and including the limit. The same is true of the integral on the left. The term $f(\epsilon)\, J_{m+1}(k_r \epsilon)$ will therefore also have a unique and finite limit, and, by the hypothesis (ii), this limit will be zero.

Thus we get

$$k_r \int_0^1 J_m(k_r x)\, x f(x)\, dx = f(1)\, \frac{m+H}{k^{\frac{3}{2}}} \{\sqrt{k_r}\, J_m(k_r)\} + \int_0^1 g(x)\, x^{\frac{1}{2}} J_{m+1}(k_r x)\, dx.$$

But
$$J_{m+1}(k_r x) = \frac{m}{k_r x} J_m(k_r x) - J'_m(k_r x).$$

Therefore

$$\sqrt{(\tfrac{1}{2}\pi k_r x)}\, J_{m+1}(k_r x)$$

$$= \sqrt{(\tfrac{1}{2}\pi k_r x)} \{(m-\tfrac{1}{2})\, J_m(k_r x)/k_r x - J'_m(k_r x) - J_m(k_r x)/2k_r x\}$$

$$= -\sin \{\tfrac{1}{4}(2m+1)\, \pi - k_r x\} + O(k_r^{-1}),$$

by formulæ (viii) and (ix). Therefore

$$\int_\epsilon^1 g(x)\, x^{\frac{1}{2}} J_{m+1}(k_r x)\, dx = -k_r^{-\frac{1}{2}} \int_\epsilon^1 g(x) \sin \{\tfrac{1}{4}(2m+1)\, \pi - k_r x\}\, dx + O(k_r^{-\frac{3}{2}}),$$

where the integral on the right vanishes, when $r \to \infty$, by the theorem of Riemann-Lebesgue. Therefore

$$\int_0^1 g(x)\, x^{\frac{1}{2}} J_{m+1}(k_r x)\, dx = k^{-\frac{1}{2}} (\eta_\epsilon + \eta_r),$$

where η_ϵ is as small as we please, provided ϵ is as small as is convenient, and η_r is as small as we please, if r is sufficiently large.

We thus get

$$\int_0^1 J_m(k_r x)\, x f(x)\, dx = O(k^{-\frac{1}{2}}) + k^{-\frac{3}{2}}(\eta_\epsilon + \eta_r)$$
$$= o(k^{-\frac{3}{2}}),$$

which proves our theorem.

§§ 36–38.

Appendix on the Convergence of the Series $\Sigma A_r J_m(k_r x)$ *at the Point* $x = 1$.

36. We state first three auxiliary theorems:—

THEOREM 1.—*The series*

$$x^{\frac{1}{2}} \Sigma C_r J_m(k_r x)\, J_m(k_r)$$

oscillates boundedly for all values of x *in the interval* $(0, 1-\epsilon)$.

[For $\eta < x < 1-\epsilon$, this theorem is obvious, using the asymptotic expression (viii) for $J_m(k_r x)$.]

THEOREM 2.—*The series*

$$\Sigma k_r^{-1} C_r J_{m+1}(k_r x)\, J_m(k_r)$$

converges boundedly for all values of x *in* $(1-\epsilon, 1)$, *converges uniformly to zero in the interval* $(\eta, 1-\epsilon)$, *and converges to unity at the point* $x = 1$.

THEOREM 3.—*The series*

$$\frac{1}{H-m} \Sigma C_r \frac{J_m(k_r x)\, J_{m+1}(k_r)}{k_r} = \Sigma C_r \frac{J_m(k_r x)\, J_m(k_r)}{k_r^2}$$

converges uniformly for all values of x *in* $(0, 1)$ *to the value* $x^m/(H-m)$.

Theorem 3 has been already proved in § 8.

To prove the other two theorems we have only to slightly modify the expression to be considered, and the procedure to be followed.

Thus to prove Theorem 1, we have plainly to introduce t^2 into the numerator of the integrand of our complex integral. The sum of n terms of our series in Theorem 1 is accordingly the value of

$$\frac{1}{2\pi i} \int \frac{t J_m(xt)}{t J'_m(t) + H J_m(t)}\, dt = \frac{1}{2\pi i} \int Q(t)\, dt,$$

say, taken round a contour which encloses the roots $\pm k_r$, ($r = 1$ to $r = n$) of the equation

$$\frac{tJ'_m(t)}{J_m(t)} + H = 0.$$

As we only require to prove this sum a bounded function of n and x, in the interval $(0 \leqslant x \leqslant 1-\epsilon)$, we shall omit numerical factors whenever convenient. Take as the contour a rectangle, containing the origin, of which the corners* are the four points $\pm\sigma \pm i\tau$. Then since $Q(t) = Q(-t)$, and $Q(u+iv)$ and $Q(u-iv)$ agree in their real parts, we see that

$$\Sigma J_m(k_r x)\, J_m(k_r) = \mathfrak{R}\left[\frac{1}{2\pi}\int_0^\tau Q(\sigma+it)\,dt - \frac{1}{2\pi i}\int_0^\sigma Q(s+i\tau)\,ds\right].$$

We choose $$\sigma = \kappa + n\pi - \tfrac{1}{2}\pi,$$

n being so large that $n^{-1}\alpha(n)$, in the expression $\kappa + n\pi + n^{-1}\alpha(n)$ for k_n/π, is numerically less than $\frac{1}{2}\pi$.

We then get, putting $\beta = \sigma + it$,

$$(2\pi\beta)^{\frac{1}{2}}\left\{J'_m(\beta) + \frac{H}{\beta}J_m(\beta)\right\} > \tfrac{1}{2}e^t. \tag{1}$$

If, on the other hand, we take $\beta = s + i\tau$, where τ is fixed and sufficiently great, and s varies from 0 to σ, we have

$$(2\pi\beta)^{\frac{1}{2}}\left\{J'_m(\beta) + \frac{H}{\beta}J_m(\beta)\right\} > \tfrac{1}{4}e^\tau. \tag{2}$$

But $$|\beta^{\frac{1}{2}}x^{\frac{1}{2}}J_m(\beta x)| < Ne^{t\alpha},$$

where $\beta = s+it$, N is independent of β and x, and $0 \leqslant x \leqslant 1$. Hence

$$|Q(\sigma+it)| < 2Ne^{tx}/e^t,$$

and therefore

$$\left|x^{\frac{1}{2}}\int_0^\tau Q(\sigma+it)\right| < 2N\int_0^\tau e^{-t(1-x)}\,dt < 2N\,\frac{1-e^{-\tau(1-x)}}{1-x},$$

* We here follow Hobson, *loc. cit.* But we require only a special case of the general result proved by him.

and is therefore bounded when we make τ sufficiently large. Similarly

$$| Q(s+i\tau) | < 4Ne^{\tau x}/e^{\tau} < 4Ne^{-\tau(1-x)},$$

and

$$x^{\frac{1}{2}} \left| \int_0^{\sigma} Q(s+i\tau)\, ds \right| < 4N\sigma e^{-\tau(1-x)}.$$

Now the roots k_r are all on the real axis. Hence there is nothing to prevent our always choosing our contour so that $\sigma e^{-\tau(1-x)}$ is as small as we please, without altering the roots enclosed.

It is plain then that Theorem 1 is true.

37. To prove Theorem 2, we have to further modify our complex integral so that it becomes

$$\frac{1}{2\pi i} \int \frac{J_{m+1}(xt)\, dt}{tJ'_m(t) + HJ_m(t)} = \frac{1}{2\pi i} \int \frac{R(t)}{t}\, dt.$$

Evidently then similar reasoning applies. We can use (1) and (2) as they stand. Also, as before,

$$| R(\sigma+it) | < 2N_1 e^{tx}/e^{t} < 2N_1 e^{-t(1-x)};$$

and therefore

$$\mathfrak{R}\left[\int_0^{\tau} \frac{R(\sigma+it)}{\sigma+it}\, dt\right] \leqslant \int_0^{\tau} | R(\sigma+it) | \frac{dt}{\sqrt{(\sigma^2+t^2)}} \leqslant 2N_1 \int_0^{\tau} \sigma^{-1}\, dt$$

$$\leqslant 2N_1 \tau/\sigma_1 \leqslant 2N_1,$$

taking $\sigma = \tau$, whether $x = 1$ or not.

On the other hand

$$\int_0^{\sigma} \frac{R(s+i\tau)}{s+i\tau}\, ds \leqslant \int_0^{\sigma} \left| \frac{R(s+i\tau)}{s+i\tau} \right| ds \leqslant \int_0^{\sigma} \frac{|R(s+i\tau)|}{\sqrt{(s^2+\tau^2)}}\, ds \leqslant \frac{4N\sigma}{\tau};$$

and is therefore bounded when we put $\sigma = \tau$.

It remains to prove the other statements of Theorem 2. This may be done as follows.

To prove that, for $\eta \leqslant x \leqslant 1-\epsilon$,

$$\Sigma k^{-1} C_r J_m(k_r)\, J_{m+1}(k_r x) \to 0$$

uniformly. First, it plainly converges uniformly, as we see by substituting the asymptotic expression (viii), and so getting the sum of series of the type

$$\Sigma p^{-1} \begin{matrix}\sin \\ \cos\end{matrix} px.$$

Denote its sum by $h(x)$. Multiplying by x^{-m} and integrating term-by-term between x_1 and x_2, we get

$$\int_{x_1}^{x_2} g(x)\, x^{-m}\, dx = \left[\Sigma k_r^{-1} C_r J_m(k_r) x^{-m} J_m(k_r x)\right]_{x_1}^{x_2} = 0,$$

using (1).

Since this is true for all pairs of values x_1 and x_2 in $\eta \leqslant x \leqslant 1-\epsilon$, and $g(x)$ is continuous,

$$g(x) = 0.$$

Thus our series converges uniformly to zero.

Again, when $x = 1$, it takes the form

$$\Sigma k_r^{-1} C_r J_m(k_r)\, J_{m+1}(k_r) = 1,$$

by Theorem 3.

38. These auxiliary theorems enable us to prove quite simply the following main theorem on the behaviour of a Bessel series at $x = 1$.

THEOREM 4 (Hobson's Theorem).—*If $f(x)$ has bounded variation at $x = 1$, and $\sqrt{x}\, f(x)$ is summable*

$$\Sigma C_r J_m(k_r) \int_0^1 J_m(k_r x)\, x\, f(x)\, dx = f(1-0).$$

First we know that, in $(\eta, 1-\epsilon)$,

$$\Sigma C_r J_m(k_r)\, J_m(k_r x)$$

oscillates boundedly, and further that

$$\Sigma \int_0^x x^m C_r J_m(k_r)\, J_m(k_r x)\, dx = \Sigma x^m k_r^{-1} C_r J_m(k_r)\, J_{m+1}(k_r x)$$

converges uniformly to zero in $(\eta, 1-\epsilon)$.

Hence, by a known theorem,* since $x^{1-m}f(x)$ is summable in $(\eta, 1-\epsilon)$,

$$\Sigma \int_{\eta}^{1-\epsilon} x^m x^{1-m} C_r J_m(k_r)\, J_m(k_r x)\, dx = 0,$$

that is,

$$\Sigma\, C_r J_m(k_r) \int_{\eta}^{1-\epsilon} x f(x)\, J_m(k_r x)\, dx = 0.$$

Secondly, we know that, in $(0, \eta)$,

$$\sqrt{x}\, \Sigma\, C_r J_m(k_r)\, J_m(k_r x)$$

oscillates boundedly (Theorem 1).

Therefore by a known theorem† this series is semi-integrable above and below, and therefore, by the theorem just quoted, if $L(x)$ and $U(x)$ be the upper and lower functions of the series,

$$\int_0^{\eta} L(x)\, g(x)\, dx \leqslant \Sigma \int_0^{\eta} C_r J_m(k_r)\, J_m(k_r x)\, x^{\frac{1}{2}} g(x)\, dx \leqslant \int_0^{\eta} U(x)\, g(x)\, dx,$$

provided $g(x)$ is summable.

Hence we may write $x^{\frac{1}{2}}f(x)$ for $g(x)$, and therefore, if B be the upper bound of the values of the functions $|L(x)|$ and $|U(x)|$ in $(0, \eta)$,

$$\left| \Sigma \int_0^{\eta} C_r J_m(k_r)\, J_m(k_r x) x\, f(x)\, dx \right| \leqslant B \int_0^{\eta} x^{\frac{1}{2}} f(x)\, dx,$$

and is therefore as small as we please, if η be small enough.

Finally, we know that, in $(1-\epsilon, 1)$,

$$\Sigma\, k_r^{-1} C_r J_m(k_r)\, J_{m+1}(k_r x)$$

converges boundedly (Theorem 2). We may therefore integrate with respect to the function of bounded variation $x^{-m}f(x)$ in $1-\epsilon$ to 1, and the result, by Theorem 2, will be zero when the interval of integration does not include the point $x = 1$, and when it does include this point will be $\{f(1)-f(1-0)\}$.

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* See my paper "Successions of Integrals and Fourier Series", *Proc. London Math. Soc.*, Ser. 2, Vol. 11, p. 64 (1912).

† See my paper "Semi-Integrals etc.", *Proc. London Math. Soc.*, Ser. 2, Vol. 9, p. 306 (1910).

Now

$$\int_{1-\epsilon}^{1} J_m(k_r x) x f(x)\, dx$$

$$= \int_{1-\epsilon}^{1} x^{-m} f(x)\, d\left\{k_r^{-1} x^{1+m} J_{m+1}(k_r x)\right\}$$

$$= \left[k_r^{-1} x f(x) J_{m+1}(k_r x)\right]_{1-\epsilon}^{1} - \int_{1-\epsilon}^{1} k_r^{-1} x^{1+m} J_{m+1}(k_r x)\, d\left\{x^{-m} f(x)\right\}.$$

Multiplying both sides by $C_r J_m(k_r x)$ and summing and using the result just given, as well as Theorem 2, we get

$$\Sigma \int_{1-\epsilon}^{1} C_r J_m(k_r) J_m(k_r x) x f(x)\, dx = f(1) - \left\{f(1) - f(1-0)\right\} = f(1-0).$$

which proves the theorem.